```
import pandas as pd

data = {
    'Student':['A','B','C','D','E'],
    'Math':[80,70,60,90,55],
    'Science':[78,68,58,88,52],
    'English':[82,72,62,92,58],
    'Computer':[85,75,65,95,60],
    'Result':['Pass','Pass','Fail','Pass','Fail']
}

df = pd.DataFrame(data)
df
```

	Student	Math	Science	English	Computer	Result
0	A	80	78	82	85	Pass
1	B	70	68	72	75	Pass
2	C	60	58	62	65	Fail
3	D	90	88	92	95	Pass
4	E	55	52	58	60	Fail

Next steps: Generate code with df New interactive sheet

4 numerical features

```
X = df[['Math','Science','English','Computer']]
y = df['Result']

print(y)
```

```
0    Pass
1    Pass
2    Fail
3    Pass
4    Fail
Name: Result, dtype: object
```

1 categorical target variable

```
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

lda = LinearDiscriminantAnalysis(n_components=1)
```

```
X_lda = lda.fit_transform(X, y)

print(X_lda)
```

```
[[ 1.09544512e+00]
 [-1.72084569e-15]
 [-1.09544512e+00]
 [ 2.19089023e+00]
 [-2.19089023e+00]]
```

```
lda_df = pd.DataFrame(X_lda, columns=['LDA_Component'])
lda_df['Result'] = y

lda_df
```

	LDA_Component	Result
0	1.095445e+00	Pass
1	-1.720846e-15	Pass
2	-1.095445e+00	Fail
3	2.190890e+00	Pass
4	-2.190890e+00	Fail

Next steps: Generate code with `lda_df` New interactive sheet